A Survey Report on

Emerging Trends in Low-Code Application Development

Prepared by: Roshan Prabhu V

Register Number: 241401084

Department: Computer Science and Business Systems

1. Introduction

Software development has evolved from hand-coded, sequential engineering toward increasingly abstracted and automated paradigms, from fourth-generation and rapid-application-development tools of the 1990s to today's cloud-based low-code platforms. This shift has accelerated as organizations pursue digital transformation under competitive pressure and as a persistent shortage of professional developers drives growth in citizen development, the building of applications by users with limited formal programming training. Vendors have further reinforced this trend by embedding generative AI copilots and natural-language application generation directly into their design environments. This survey reviews ten leading commercial low-code platforms individually, followed by a comparative analysis and a review of emerging trends, with scope limited to general-purpose enterprise low-code platforms.

2. Survey of Major Low-Code Platforms

2.1 Microsoft Power Apps

Microsoft Power Apps was introduced in 2016 as part of the Microsoft Power Platform, alongside Power Automate, Power BI, and Power Virtual Agents. Developed and maintained by Microsoft, it has grown from a departmental app builder into a core component of Microsoft's enterprise digital transformation strategy. Power Apps targets both citizen developers and professional developers, supports canvas apps (pixel-level custom layouts) and model-driven apps (generated from a Dataverse data model), and is deployed as a cloud-first SaaS platform tightly coupled to the Microsoft 365 and Azure ecosystem. It currently holds one of the largest market shares among enterprise low-code platforms, driven by bundling within Microsoft 365 and Dynamics 365 licenses.

Key Features:

- Visual Development: canvas and model-driven design surfaces with drag-and-drop layout
- AI Features: Copilot for natural-language app generation; AI Builder for prebuilt/custom ML models
- API Integration: over 1,000 prebuilt connectors and custom connector support
- Security: Microsoft Entra ID (Azure AD) integration, role-based access, data loss prevention policies

Advantages:

- Deep, native integration with Microsoft 365 and Dynamics 365
- Relatively gentle learning curve for existing Microsoft users
- Strong generative-AI (Copilot) support for app scaffolding
- Cost-effective for organizations already holding Microsoft licenses

Disadvantages:

- Complex, multi-tiered licensing that can be difficult to predict at scale
- Canvas apps can face performance limits with very large datasets
- Governance can be difficult without a mature Center of Excellence practice

Typical Use Cases: Healthcare (patient intake forms), Banking (expense and approval workflows), Retail (inventory and store operations apps), Manufacturing (inspection and quality checklists), Education (administrative workflow apps), Government (internal case tracking)

Mini Summary: Power Apps is most suitable for organizations already invested in the Microsoft ecosystem seeking to empower both citizen and professional developers with a cost-effective, AI-enhanced low-code environment for internal business applications.

2.2 OutSystems

OutSystems was founded in 2001 in Portugal and has since become one of the most recognized enterprise-grade low-code platforms, consistently positioned as a market leader by Gartner and Forrester. It targets professional development teams building complex, mission-critical applications, and is deployed via OutSystems-managed cloud infrastructure or self-managed deployment on major public clouds. OutSystems occupies a premium market position among platforms oriented toward technical depth and enterprise scalability rather than pure citizen development.

Key Features:

- Visual Development: full-stack visual modeling for UI, logic, and data with permitted custom code extension
- AI Features: AI-assisted code generation, architecture pattern analysis, AI-driven test recommendations
- API Integration: REST/SOAP integration studio and API management
- Security: role-based access control, encrypted storage, integrated vulnerability scanning

Advantages:

- Enterprise-grade scalability suited to mission-critical applications
- Compiled execution model narrows the performance gap with hand-coded software
- Strong flexibility to extend generated applications with custom code
- Consistently rated a leader in Gartner and Forrester evaluations

Disadvantages:

- Higher licensing cost relative to citizen-developer-focused platforms
- Steeper learning curve for teams new to its architecture concepts
- Less suited to very simple, low-stakes departmental applications

Typical Use Cases: Banking (core system modernization), Manufacturing (supply chain portals), Telecommunications (customer self-service apps), Healthcare (patient management systems), Insurance (claims platforms)

Mini Summary: OutSystems is best suited to large enterprises with professional development teams that need to build complex, high-performance, and highly governed applications at scale, rather than simple departmental tools.

2.3 Mendix

Mendix was founded in the Netherlands in 2005 and acquired by Siemens in 2018, positioning it distinctively at the intersection of enterprise IT and industrial digitalization. It targets both professional developers and business analysts through a strongly collaborative visual modeling environment, and supports deployment to Mendix Cloud, private cloud, or on-premises infrastructure. Its Siemens ownership has strengthened its market position in manufacturing and industrial IoT scenarios alongside general enterprise application development.

Key Features:

- Visual Development: shared, collaborative modeling canvas for business and IT users
- AI Features: Mendix Assist for contextual modeling suggestions; generative-AI application scaffolding
- API Integration: REST/OData/GraphQL connectors and integration marketplace
- Security: role-based security models, encrypted data, compliance certifications

Advantages:

- Strong collaborative modeling for mixed business/IT teams
- Flexible deployment across cloud, private cloud, and on-premises

- Distinctive strength in industrial and manufacturing digitalization via Siemens integration
- Free tier available for small applications and learning

Disadvantages:

- Node-based/consumption pricing can become expensive at scale
- Advanced customization still requires familiarity with Mendix-specific concepts
- Smaller partner ecosystem in some regions compared to Microsoft or Salesforce

Typical Use Cases: Manufacturing (production monitoring, IIoT dashboards), Finance (loan and account workflows), Logistics (warehouse and fleet applications), Government (citizen service portals), Healthcare (care coordination apps)

Mini Summary: Mendix is particularly well suited to organizations, especially in manufacturing and industrial sectors, that need strong business-IT collaboration and flexible deployment options across cloud and on-premises environments.

2.4 Appian

Appian, founded in 1999 and headquartered in the United States, originated in business process management (BPM) and has since expanded into a full low-code application platform with native robotic process automation (RPA) and process mining. It targets large enterprises building complex, long-running business processes, deployed via cloud or hybrid infrastructure. Appian is consistently positioned among market leaders for organizations requiring sophisticated case management and regulatory compliance.

Key Features:

- Visual Development: process-centric visual modeling with parallel branches and case management
- AI Features: document understanding, predictive analytics, and generative-AI process assistance
- API Integration: extensive connector library and API management tooling
- Security: enterprise-grade encryption, audit trails, and compliance certifications

Advantages:

- Best-in-class business process management and case management capability
- Native RPA reduces dependency on separate automation tools
- Strong regulatory and compliance support for regulated industries
- Scales well for large, mission-critical enterprise deployments

Disadvantages:

- Premium pricing positions it beyond reach for smaller organizations
- Less efficient for very simple, single-purpose departmental applications
- Requires process-modeling expertise to exploit its full capability

Typical Use Cases: Insurance (claims processing), Government (case management, permitting), Banking (loan and compliance workflows), Healthcare (care coordination), Manufacturing (supply chain case tracking)

Mini Summary: Appian is most suitable for large enterprises in regulated industries that require sophisticated, auditable business process automation rather than simple standalone applications.

2.5 Salesforce Lightning Platform

The Salesforce Lightning Platform extends the Salesforce customer relationship management (CRM) ecosystem, first launched in 1999, into a general-purpose low-code application platform. Owned by Salesforce, Inc., it targets organizations already using Salesforce for sales, service, or marketing, and is delivered exclusively as a multi-tenant cloud SaaS platform. It holds a leading market position specifically among CRM-centric enterprises across virtually every industry.

Key Features:

- Visual Development: Lightning App Builder and Flow Builder for declarative development
- AI Features: Salesforce Einstein predictive analytics and Einstein Copilot generative AI assistant
- API Integration: extensive REST/SOAP APIs and AppExchange marketplace connectors
- Security: Salesforce Shield encryption, field-level security, and compliance tooling

Advantages:

- Native, real-time access to Salesforce CRM data without separate integration
- Mature Einstein AI suite with strong predictive and generative capability
- Combination of declarative tools (for citizen developers) and Apex/LWC (for professional developers)
- Large global partner and developer ecosystem

Disadvantages:

- Best value is realized primarily within the Salesforce ecosystem
- Can become costly as user and feature licensing scales
- Apex/LWC development still requires genuine programming skill

Typical Use Cases: Retail (customer engagement apps), Banking (relationship management extensions), Healthcare (patient relationship tools), Telecommunications (service case management), Manufacturing (partner portals)

Mini Summary: The Lightning Platform is most suitable for organizations already committed to Salesforce CRM that want to extend customer-related processes with custom low-code applications sharing the same data model.

2.6 Google AppSheet

AppSheet was founded in 2014 and acquired by Google in 2020, after which it was integrated into the Google Workspace ecosystem. It targets business users and small teams rather than professional developers, distinguished by its ability to generate applications directly from spreadsheet data in Google Sheets, Excel, or cloud databases. AppSheet occupies a strong market position specifically among no-code, spreadsheet-driven application builders.

Key Features:

- Visual Development: automatic UI generation inferred from spreadsheet structure
- AI Features: automated format detection and natural-language-assisted app generation
- API Integration: webhook and API-based integration with external services
- Security: Google Workspace identity, role-based row-level security

Advantages:

- Extremely fast application creation directly from existing spreadsheets
- Strong offline mobile capability with automatic synchronization
- Very low barrier to entry for non-technical business users
- Free tier available for individual and small-team use

Disadvantages:

- Limited capability for complex, multi-step business process workflows
- Less suited to large-scale enterprise governance requirements
- AI and automation features are less mature than enterprise-oriented platforms

Typical Use Cases: Logistics (delivery and inspection tracking), Retail (inventory and store checklists), Healthcare (field visit tracking), Education (attendance and survey apps), Manufacturing (equipment inspection)

Mini Summary: Google AppSheet is best suited to business users and small teams needing to quickly convert spreadsheet-based processes into mobile-friendly applications, particularly for field operations requiring offline access.

2.7 Zoho Creator

Zoho Creator was launched in 2006 by Zoho Corporation, an India-based enterprise software vendor known for affordable business applications. It targets small and medium-sized enterprises (SMEs) that lack dedicated IT departments, and is delivered as a cloud SaaS platform with a moderate scripting layer (Deluge) for customization. Zoho Creator holds a strong market position specifically within the cost-conscious SME segment.

Key Features:

- Visual Development: drag-and-drop form and page builder with prebuilt templates
- AI Features: Zia AI assistant for prediction and anomaly detection
- API Integration: REST API support and Zoho ecosystem connectors
- Security: role-based access control and standard data encryption

Advantages:

- Very cost-effective licensing suited to SME budgets
- Simple, approachable interface requiring minimal training
- Tight integration with the broader, affordable Zoho suite
- Fast deployment for straightforward departmental applications

Disadvantages:

- Limited enterprise-scale governance and audit capability
- AI features are less extensive than platforms like Salesforce or Power Apps
- Smaller partner and integration ecosystem than major enterprise vendors

Typical Use Cases: Retail (inventory and point-of-sale support), Manufacturing (light production tracking), Education (administrative tools), Healthcare (small-clinic scheduling), Logistics (small-fleet tracking)

Mini Summary: Zoho Creator is most suitable for small and medium enterprises seeking affordable, straightforward digitization of core business processes without the complexity or cost of enterprise-grade platforms.

2.8 ServiceNow App Engine

ServiceNow App Engine extends the ServiceNow platform, founded in 2004 and historically centered on IT service management (ITSM), into a broader low-code development environment. It targets large enterprises, particularly those with mature ServiceNow deployments, and is delivered as a cloud SaaS platform with strong built-in governance. ServiceNow holds a leading market position among enterprises seeking IT-governed citizen development.

Key Features:

- Visual Development: App Engine Studio with guided, template-driven application creation
- AI Features: Now Assist generative-AI capabilities for app creation and workflow suggestions
- API Integration: REST/SOAP APIs and a large integration hub of prebuilt connectors
- Security: enterprise-grade access control, encryption, and compliance certifications

Advantages:

- Strong, distinctive centralized governance over citizen-developed applications
- Deep native integration with IT service management data and processes

- Reliable enterprise-grade security and compliance posture
- Reduces shadow IT risk through built-in review and approval workflows

Disadvantages:

- Primarily valuable to organizations already using the ServiceNow platform
- Licensing cost can be significant for smaller organizations
- Less flexible for applications unrelated to service or workflow management

Typical Use Cases: Telecommunications (network operations workflows), Banking (internal service requests), Healthcare (compliance and incident workflows), Government (case and service management), Manufacturing (asset and maintenance workflows)

Mini Summary: ServiceNow App Engine is most suitable for large enterprises already using ServiceNow that need strong centralized governance over citizen-developed applications extending IT service management practices.

2.9 Oracle APEX

Oracle Application Express (APEX), first released in 2004, is a low-code development platform built directly into the Oracle Database. Developed and maintained by Oracle Corporation, it targets developers and data teams already working within an Oracle Database environment, and is distinctive among the platforms surveyed for its database-centric architecture rather than a separate application server model. APEX holds a strong market position among Oracle-oriented enterprises and is offered at no additional license cost to Oracle Database customers.

Key Features:

- Visual Development: Page Designer with drag-and-drop components bound directly to database objects
- AI Features: emerging AI-assisted development within Oracle Cloud tooling
- API Integration: RESTful services generated directly from database objects
- Security: Oracle Database security model, including row-level security and auditing

Advantages:

- Extremely tight, high-performance integration with Oracle Database
- No additional licensing cost for existing Oracle Database customers
- Mature, stable platform with two decades of production use
- Low infrastructure overhead since it runs within the database environment

Disadvantages:

- Value proposition is largely limited to organizations already using Oracle Database
- AI and generative-AI capabilities are less mature than newer cloud-native platforms
- User interface styling is less contemporary than some competing platforms

Typical Use Cases: Banking (core data-centric applications), Government (regulatory reporting systems), Manufacturing (production data applications), Insurance (policy administration on Oracle data), Telecommunications (network data applications)

Mini Summary: Oracle APEX is most suitable for organizations with substantial existing Oracle Database investment that need to rapidly build secure, data-centric applications directly on top of that infrastructure.

2.10 SAP Build

SAP Build is SAP's unified low-code and process-automation suite within the SAP Business Technology Platform (BTP), consolidating earlier tools such as SAP AppGyver and SAP Process Automation into a single environment introduced in 2023. Developed and maintained by SAP SE, it targets both professional developers and business users inside large enterprises already running SAP ERP or S/4HANA, and is delivered as a cloud SaaS platform on SAP BTP. SAP Build holds a distinctive market position among enterprises with deep existing SAP investment, competing less on breadth of citizen-developer adoption and more on native depth of SAP process and data integration.

Key Features:

- Visual Development: SAP Build Apps drag-and-drop UI builder plus SAP Build Process Automation for workflows
- AI Features: Joule generative-AI assistant for app and workflow generation
- API Integration: SAP Business Accelerator Hub connectors and Open Connectors for third-party systems
- Security: SAP Identity Authentication Service and role-based access aligned with SAP authorization concepts

Advantages:

- Unmatched native integration with SAP ERP, S/4HANA, and SAP data models
- Joule generative AI accelerates both app and workflow creation
- Combines UI app-building and process automation in a single suite
- Runs on a multi-cloud-capable platform (BTP) rather than a single hyperscaler

Disadvantages:

- Value proposition is largely limited to organizations already running SAP
- Smaller general-purpose citizen-developer community than Power Apps or AppSheet
- Licensing is typically bundled with broader, costly SAP BTP consumption plans

Typical Use Cases: Manufacturing (procurement and supply chain workflow extensions), Retail (order and inventory process automation), Public Sector (SAP-based case and approval workflows), Finance (invoice and financial approval automation)

Mini Summary: SAP Build is most suitable for large enterprises with substantial existing SAP ERP or S/4HANA investment that need to extend and automate SAP-centric business processes without leaving the SAP ecosystem.

3. Comparative Analysis

Building on the individual platform surveys above, this section compares all ten platforms using structured tables and charts rather than repeating platform descriptions.

Table 1. General Comparison

Platform	Vendor	Deployment	AI Support	Mobile Support	Best For	Scalability
Power Apps	Microsoft	Cloud/Hybrid	Copilot, AI Builder	Native	Microsoft-centric orgs	High
OutSystems	OutSystems	Cloud/On-prem	AI code & arch. analysis	Native	Complex enterprise apps	Very High
Mendix	Siemens	Cloud/Private/On-prem	Mendix Assist	Native, offline	Industrial/manufacturing	High
Appian	Appian Corp.	Cloud/Hybrid	Process AI, doc. AI	Native	Complex BPM/case mgmt.	Very High
Salesforce Lightning	Salesforce	Cloud	Einstein, Copilot	Native	CRM-centric orgs	Very High
Google AppSheet	Google	Cloud	AI format detection	Native, offline	Spreadsheet-driven apps	Moderate
Zoho Creator	Zoho Corp.	Cloud	Zia (limited)	Native	SMEs	Moderate
ServiceNow App Engine	ServiceNow	Cloud	Now Assist AI	Native	IT-governed enterprises	Very High
Oracle APEX	Oracle	Cloud/On-prem (DB-based)	Emerging	Responsive	Oracle DB-centric orgs	High
SAP Build	SAP SE	Cloud (SAP BTP)	Joule AI assistant	Native/Responsive	SAP ERP-centric orgs	High

Table 2. Feature Comparison (Excellent / Good / Average / Limited)

Capability	Power Apps	OutSystems	Mendix	Appian	Salesforce	AppSheet	Zoho	ServiceNow	APEX	SAP Build
Visual Development	Excellent	Excellent	Excellent	Good	Good	Excellent	Good	Good	Good	Good
Workflow	Good	Excellent	Excellent	Excellent	Good	Average	Average	Excellent	Good	Excellent
AI Assistant	Excellent	Good	Good	Good	Excellent	Average	Average	Good	Limited	Good
API Integration	Excellent	Excellent	Good	Good	Excellent	Average	Average	Good	Good	Good
Cloud Support	Excellent	Excellent	Good	Good	Excellent	Good	Good	Excellent	Good	Excellent
Database Support	Good	Good	Good	Good	Good	Average	Average	Average	Excellent	Excellent
Security	Excellent	Excellent	Good	Excellent	Excellent	Average	Average	Excellent	Excellent	Excellent
DevOps	Good	Excellent	Good	Good	Good	Limited	Limited	Good	Average	Good
Citizen Developer Support	Excellent	Average	Good	Average	Good	Excellent	Excellent	Good	Average	Average
Offline Capability	Average	Average	Good	Limited	Limited	Excellent	Limited	Limited	Limited	Limited
Analytics	Good	Average	Average	Good	Excellent	Average	Average	Good	Good	Good

Table 3. Advantages and Limitations

Platform	Major Advantages	Major Limitations
Power Apps	Microsoft 365 integration; strong Copilot AI	Complex licensing; canvas app limits
OutSystems	Enterprise scalability; architecture governance	Higher cost; steep learning curve
Mendix	Collaboration; flexible deployment	Node-based pricing can escalate
Appian	Best-in-class BPM/case management	Premium pricing; process expertise needed
Salesforce Lightning	Native CRM data; mature AI (Einstein)	Best value within Salesforce ecosystem
Google AppSheet	Fast spreadsheet-to-app; offline mobile	Limited complex workflow capability
Zoho Creator	Very cost-effective for SMEs	Limited enterprise governance/AI depth
ServiceNow App Engine	Strong governance; ITSM integration	Value mainly for existing ServiceNow users
Oracle APEX	Tight Oracle DB integration; low extra cost	Value mainly for existing Oracle DB users
SAP Build	Unmatched SAP ERP/S4HANA integration; Joule AI	Value mainly for existing SAP customers

Table 4. Industry Adoption (High / Moderate / Low)

Industry	Power Apps	OutSystems	Mendix	Appian	Salesforce	ServiceNow	APEX	SAP Build
Healthcare	High	Moderate	Moderate	High	Moderate	Moderate	Moderate	Low
Finance	Moderate	High	Moderate	High	High	Moderate	High	Moderate
Manufacturing	High	High	High	Moderate	Moderate	Moderate	Moderate	High
Education	Moderate	Low	Low	Low	Moderate	Low	Low	Low
Government	Moderate	High	Moderate	High	Moderate	High	High	Moderate
Retail	High	Moderate	Moderate	Low	High	Low	Low	High
Telecommunications	Moderate	High	Moderate	Moderate	Moderate	High	Moderate	Moderate

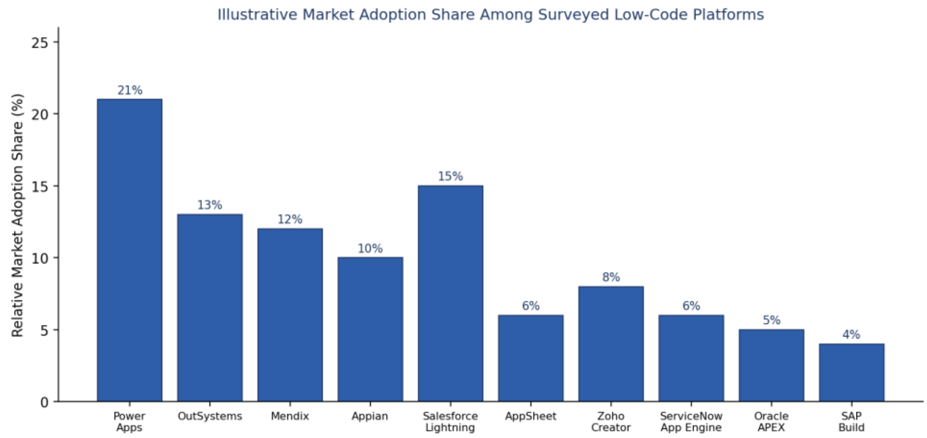

Figure 11. Chart 1 — Enterprise Adoption of Low-Code Platforms (Illustrative).

4. Emerging Trends in Low-Code Development

Artificial intelligence is now the dominant force reshaping low-code development. Generative AI and embedded AI copilots translate natural-language descriptions into working application scaffolding and offer real-time, context-aware suggestions during development, while natural language application development lets business users describe functionality conversationally. Agentic AI and autonomous application generation push this further toward AI agents that independently plan, build, test, and refine application components with minimal human intervention, and AI-assisted testing and deployment are automating quality assurance and release-risk prediction.

A parallel set of trends concerns automation and architecture. Hyperautomation combines low-code, robotic process automation, business process management, and AI into a unified fabric addressing entire business processes rather than isolated applications, while intelligent process automation embeds continuous learning so workflows adapt without manual reconfiguration. Cloud-native low-code platforms, built on containers and microservices, and composable applications assembled from reusable business capabilities are improving scalability and long-term agility, while multi-experience development, edge computing, and IoT integration extend consistent experiences across web, mobile, and distributed operational environments.

Governance and trust are maturing alongside these capabilities. DevSecOps integration embeds automated security scanning directly into deployment pipelines, enterprise governance frameworks provide centralized application inventories and approval workflows to manage citizen-development risk, and explainable AI is emerging to validate the reasoning behind AI-generated logic in regulated industries. Direct integration of large language models into low-code platforms is accelerating across nearly all vendors and is likely to become the primary interface for low-code development within a few years, further blurring the line between low-code and autonomous software engineering, as reflected in the adoption levels shown below.

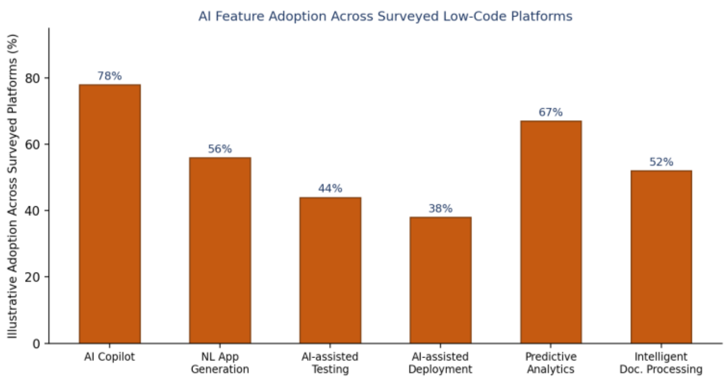

Figure 13. AI Feature Adoption Across Surveyed Low-Code Platforms.

5. Conclusion

This survey has reviewed ten major low-code development platforms, comparing their features, advantages, and industry adoption. Platform suitability depends on context: OutSystems and Appian suit large, governed enterprise applications; Power Apps, Salesforce, Oracle APEX, and SAP Build suit organizations embedded in their ecosystems; and Zoho Creator and AppSheet best serve small teams seeking cost-effective digitization. The market is moving toward deeper AI integration and maturing governance.